
Learning Skill-Attributes for Transferable Assessment in Video

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Abstract

Skill assessment from video entails rating the quality of a person’s physical performance and explaining what could be done better. Today’s models specialize for an individual sport, and suffer from the high cost and scarcity of expert-level supervision across the long tail of sports. Towards closing that gap, we explore transferable video representations for skill assessment. Our CROSSTRAINER approach discovers *skill-attributes*—such as balance, control, and hand positioning—whose meaning transcends the boundaries of any given sport, then trains a multimodal language model to generate actionable feedback for a novel video, e.g., “*lift hands more to generate more power*” as well as its proficiency level, e.g., *early expert*. We validate the new model on multiple datasets for both cross-sport (transfer) and intra-sport (in-domain) settings, where it achieves gains up to 60% relative to the state of the art. By abstracting out the shared behaviors indicative of human skill, the proposed video representation generalizes substantially better than an array of existing techniques, enriching today’s multimodal large language models. Project page: <https://vision.cs.utexas.edu/projects/CrossTrainer/>.

1 Introduction

The basis of assessing skilled physical activities, particularly sports, is largely visual. Precisely how a tennis player grasps and swings their racquet; how a basketball player releases the ball to shoot a free throw; how a rock climber stretches and pulls to traverse the boulder—such visual details are discernible to the expert eye and essential for providing meaningful coaching. Advances in multimodal video understanding could, therefore, transform AI-assisted coaching and skill assessment. For example, future AI agents could provide personalized feedback to users based on videos captured on their phone or smartglasses, greatly expanding the accessibility of 1-1 coaching. Similarly, AI could analyze how multiple players’ skills would complement each other when building a team, or even detect patterns in injuries as a function of execution style.

Towards achieving the above, the core vision and machine learning task is as follows: given a video of an athlete’s performance, estimate their skill level [12, 28, 37, 39, 67, 70, 71, 73] and indicate what could be improved [9, 16, 68].

This task raises important unsolved challenges. First of all, assessing skill requires *fine-grained* understanding of all physical aspects. Whereas traditional action understanding emphasizes high-level semantics [17, 18, 27, 55, 60, 87, 88, 92]—and thus seeks *invariance* to execution differences—here the subtle differences are exactly what matters. For example, a novice and expert shooting a soccer ball into the net might accomplish the same overall action, but details about their approach, footwork, and trajectory of the ball are essential to analyze skill. Secondly, compounding the challenge, supervision is costly and difficult to scale across the *long tail* of sports. An estimated 8,000 unique sports exist today across the world, but only a small subset is widely recorded and distributed—often driven by geographic and economic factors. Similarly, expert supervision is expensive and cannot be scaled up easily using traditional annotator pools. Making it worse, due to the common assumption that the axes of evaluation vary so wildly between sports, all prior work trains and tests on *in-domain* data, i.e., the same drill, exercise, or sport [9, 12, 37, 70, 70–72, 85, 98].



Figure 1: **Overview of the idea.** Given a short video of an athletic skill, what could be better? Given video demonstrations from multiple sports, we learn skill-attributes that are incorrectly demonstrated, e.g., wrong foot positioning in badminton (top). These skill-attributes are common across various sports, and transfer to novel uncommon sports, e.g., shinty (bottom). Our method improves both the in-domain and zero-shot settings. Sports chosen for illustration; see Sec. 4 for dataset details.

We propose CROSSTRAINER, a new approach to video-based skill analysis that accounts for these challenges. Our key insight is to learn a universal *skill-attribute* representation that transcends the boundaries of sports and hence allows sharing and transfer between them. A skill-attribute is a describable fine-grained concept about the physical performance that can take on different visual forms in different sports, e.g., *ball control*, *foot positioning*, *coordination*, *timing*, or *balance*. By automatically learning these shared properties from video-language commentaries, and then surfacing them during training, we aim to amplify the value of the limited training data available for any one sport.

Building on the skill-attributes, we then train a generative multimodal language model to predict i) which skill-attributes are incorrectly demonstrated in a given video, ii) what the proficiency level is, and iii) free-form commentary about which adjustments would improve the performance. Essentially we factor skill assessment into its generalizable cross-sport component (e.g., *lacks control*) and its focused sports-specific component (e.g., “*you should increase the spacing between your legs for better control of the soccer ball*”). Our formulation aims to unlock the translation of skill assessment from widespread, highly marketed sports to uncommon, low-resource sports, such as *kabaddi*, *frisbee*, *waterpolo*, *shinty*, *kho-kho*, or *bandy*. See Fig. 1.

Though this represents a departure from today’s AI models [9, 12, 37, 70, 70–72, 85, 98], cognitive science supports both transferability across sports having similar skill sets and shared terminology for assessment [19, 65, 77, 82]. For example, basketball players make better decisions playing soccer compared to tennis players, due to the former two sports’ shared underlying properties [77]; similarly,

contact-sport athletes exhibit better zero-shot understanding of a new contact sport compared to non-contact sport athletes [19]; and transferable skills arise from similar affordances [43, 81] or environmental properties [82]. Our work is the first to translate these intuitive findings from cognitive science into a working video understanding model.

We validate our ideas on three diverse datasets: Ego-Exo4D [37], which contains soccer, basketball, and rock climbing; QEVD [68], which contains fitness exercises; and in-the-wild YouTube videos of people tutoring physical activities. We show that training an intermediate skill representation yields both superior *in-domain* performance (for familiar sports) as well as better *zero-shot* performance (where the particular sport or drill is never seen by the model during training). CROSSTRAINER outperforms all competitive baselines on all axes of assessment—actionable feedback, skill-attributes, and proficiency estimation—with relative gains up to 60%. Furthermore, compared to any of the baselines, CROSSTRAINER shows much more graceful degradation when transferring to a novel sport. Overall our approach is a stepping stone for learning a unified skill-centric video representation for assessment and coaching in the presence of real-world data and annotation constraints.

2 Related Work

Learning representations from videos. Substantial research explores new ways to learn video representations [6, 48, 51, 74, 92], targeting video understanding tasks of action recognition [36, 45, 49, 89], action anticipation [1, 32, 33, 35, 57], procedural understanding [13, 14, 20, 62, 101], and temporal step localization [3, 58, 97, 99, 100]. Instructional videos offer a valuable window into skilled human activity [59, 84, 103], and recent work explores how to navigate between related how-to’s [7, 8, 102] and identify their differences [61]. Whereas prior work explores activity-centric representations emphasizing the semantics of *what* is being done, our problem requires capturing *how* it is being done, which we show is essential for skill assessment.

Video-based sports analytics. Sports analytics and assessment raises a number of interesting challenges for computer vision [12, 37, 70, 71, 73, 85, 98]. Multiple ongoing workshops and challenges [22, 34, 64] offer tasks including ball spotting, foul recognition, and game state reconstruction. Prior work on skill assessment either assigns a score (or equivalence class label) to a video demonstration [12, 37, 67, 70, 71, 73], optimizes a group contrastive score distribution [85, 98], or chooses the better of two demonstrations [12, 29]. Broadening the scope of skill assessment beyond scoring videos, ExpertAF [9] aims to provide *actionable feedback* in the form of natural language commentary, relevant video retrievals, and generated pose corrections. As discussed above, all the above existing work in video-based skill assessment is limited to in-domain testing, whereas we explore transfer *across* sports, enabled by the proposed skill-attributes. In addition, orthogonal to the transfer contribution, our results across three datasets representing 6 distinct sports and fitness activities and 30 distinct drills raises the bar in breadth of validation compared to any prior skill assessment work.

Zero-shot generalization and attributes. Testing on novel classes and scenarios is crucial for real open-world settings. In one line of work, zero-shot generalization is enabled by shared multimodal representations learned from noisy vision-language data, benefiting image classification [75, 90, 93], action recognition [10, 95, 96], video to text and text to video retrieval [6, 80, 87, 88, 91, 92], or image segmentation [15, 26, 40]. In another line of work, *attributes* are intermediate variables [41, 42, 54, 66, 83, 94] that can express a new category even without training images (e.g., zebras are *black and white* and *striped*). Though they share our motivation for shared representations, all of the existing methods focus on semantics (what) as opposed to dynamic execution (how), and none are directly applicable to skill assessment from video. Ours is the first work to explore attributes for fine-grained activities in video and the first to demonstrate the relevance for skill feedback.

3 Method

We introduce the problem statement in Sec. 3.1, followed by an overview of key datasets (Sec. 3.2), our idea to extract skill-attributes (Sec. 3.3), the full model (Sec. 3.4), and training (Sec. 3.5).



Figure 2: **Discovered skill-attributes** from Ego-Exo4D [37] (left) and QEVD [68] (right). We see phrases reflecting generalizable physical concepts like control, hand/body positioning, and movement.

3.1 Problem definition

At inference time, given a short video clip $V \in \mathcal{D}_{te}$ from the test set $\mathcal{D}_{te}$, we want to assess its quality, even if the exact same skill/drill was never seen in training—and even (more extreme) if the sport in V was not seen during training. The desired assessment output covers three aspects: report the skill-attributes performed incorrectly, generate actionable feedback that can help the learner improve, and finally, estimate a proficiency score. Consistent with the transfer observed in cognitive science studies [43, 81, 82], we focus on physical skills executed by an individual and hence assume the video contains one person of interest; multi-player team interactions are also interesting but less amenable to transfer and outside the scope of this work.

To handle this task of assessing both in-domain and novel data, we employ a two-stage training process. In the pretraining stage, we train the model to generate the incorrectly demonstrated skill-attributes, i.e., we learn a function $\mathcal{F}_a$ to predict the skill-attributes $\hat{S} = \{s_1, s_2, \dots\}$ that the person in the video should improve on:

$$\mathcal{F}_a(V | \mathcal{D}_{tr}) = \hat{S}, \quad (1)$$

where $\mathcal{D}_{tr}$ denotes the training set, e.g., if a person in the video is dribbling a soccer ball, S can be *control* and *leg positioning*, supposing the person is incorrectly executing those two. See Fig. 1, left.

In the second stage, we finetune the model for the remaining two aspects of assessment. Firstly, we generate feedback conditioned on the inferred skill-attribute set $\hat{S}$,

$$\mathcal{F}_t(V, \hat{S} | \mathcal{D}_{tr}) = T, \quad (2)$$

where T is a textual actionable feedback statement for improvement, e.g., “*the player is too straight and should bend more for a better control*”. While the skill-attributes are words or short phrases indicating suboptimal dimensions in general terms, the feedback consists of sentence(s) elaborating on what to fix in the context of the observed sport. Finally, we generate the proficiency level of the person in V , again conditioned on $\hat{S}$:

$$\mathcal{F}_p(V, \hat{S} | \mathcal{D}_{tr}) = P, \quad (3)$$

where P is the proficiency class label, e.g., *novice*, *intermediate*, *early expert* or *late expert*.

All aspects of assessment are evaluated in both the *in-domain* and *zero-shot* settings, to explore both the absolute performance of our proposed approach with respect to the literature (in-domain) as well as its ability to transfer to novel sports and scenarios (zero-shot). In the experiments we tackle multiple datasets of various sports and fitness activities, introduced next and detailed more in Sec. 3.5.

3.2 Skilled physical activity datasets

Before introducing our model, we next overview the key existing datasets leveraged in this study, since being familiar with their contents will help visualize our overall learning paradigm. **Ego-Exo4D** [37] contains 2,593 videos, totaling 239 hours with 289 total participants playing 3 sports—soccer, rock climbing, and basketball. **Qualcomm Exercise Videos Dataset (QEVD)** [68] has 223 long home fitness exercise videos, totaling 13 hours, where 28 total participants perform 23 structured workout

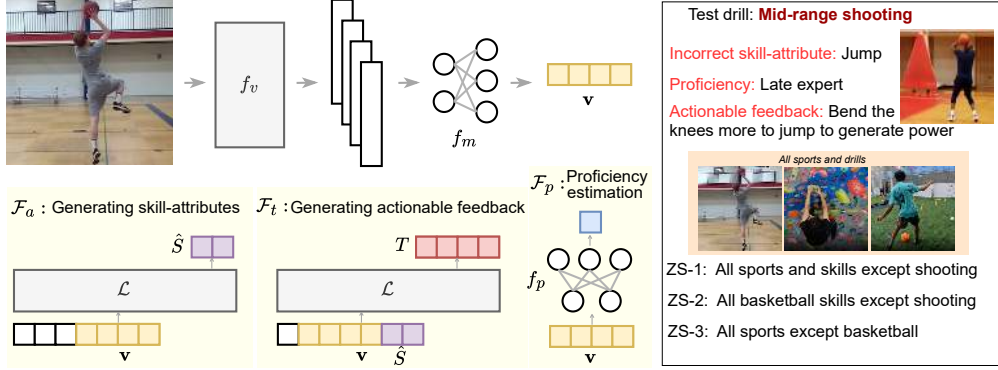


Figure 3: **Method overview and evaluation settings.** (Left) We encode videos into tokens $\mathbf{v}$ that can be fed to a multimodal LLM $\mathcal{L}$, with a mapper f_m that trains for skill-attributes. We use these visual tokens to generate skill-attributes (bottom left). Next, this pretraining is used to generate actionable feedback (bottom middle) and proficiency score (bottom right). (Right) Example of the various training settings for in-domain and zero-shot.

moves (like jumping jacks, planks, squats, etc.). To stretch our model on in-the-wild zero-shot settings, we also show qualitative results using **YouTube sports** videos collected by us—*frisbee*, *water polo* and *soccer (juggling)*.

Both Ego-Exo4D and QEVD provide actionable feedback commentary for the videos, which consists of timestamped positive and negative critiques, ideas to correct form, pacing, or other aspects of execution. Each sport is critiqued by expert coaches or players from that same sport. Given a video V , the expert pauses the video at multiple timepoints t and provides verbal feedback, e.g., “...*here he’s showing good control but lacks speed, which is critical to an effective dribble...*”. Ego-Exo4D additionally has proficiency labels corresponding to four distinct skill levels, from novice to late expert. See Figure 5 and Supp. for examples.

3.3 Stage I: Discovering skill-attributes for pretraining supervision

To supervise the training of $\mathcal{F}_a$, we obtain the skill-attributes set S for every video demonstration by sourcing it from Ego-Exo4D and QEVD commentaries—totaling around 34k unique feedback strings. We use these commentaries as a signal to extract the skill-attributes the learner in the video should improve on, as detailed next. While the Ego-Exo4D and QEVD expert commentary represent existing video-language datasets nicely suited for our purposes, such commentary could potentially be sourced “in the wild” from current video sharing platforms (e.g., Reddit, TikTok, etc.) where people provide similar verbal assessments for videos shared on social media. See Supp.

For each training sample, we prompt a large language model (LLM) [2] to extract skill-attributes that are suboptimally demonstrated according to its expert commentary text T . We provide examples to help define the intent of skill-attributes to the LLM. See Supp. for full prompt. This process yields $S = \{s_1, s_2, \dots\}$ from the expert commentary T at time t . Note that $|S|$ can be of any length. Fig. 2 shows word clouds of the discovered skill-attributes for the two video datasets. We observe that many salient phrases transcend sport boundaries, like *body positioning*, *balance*, *control*, and *movement*.

Finally, we sample a video chunk around the time t , i.e., $[t - \mu_1, t + \mu_2]$ and associate it with the skill-attribute set S .¹ By using an LLM combined with the raw commentary data, we obtain diverse, open-vocabulary skill-attributes for every training sample. We draw on these resulting LLM-inferred annotations when training the multimodal LLM (Sec. 3.4).

3.4 Stage II: Skill assessment from video

Next we present our model design for learning $\mathcal{F}_a$. Taking inspiration from the success of multimodal LLMs [44, 46, 53, 56], we convert a query video V into visual tokens. Next, we input the visual

¹The commentary text has only a timepoint t , not a temporal extent; this window extraction is consistent with how the temporal ambiguity is handled in prior work [9, 51, 76].

tokens to a multimodal LLM, along with a prompt to predict the skill-attributes S . The output is parsed to create the prediction $\hat{S}$. Fig. 3 shows the overview, and we describe each step next.

Encoding video demonstrations. We first use a standard video encoder f_v (we use EgoVLP2 [74] and CLIP [75], though others are possible) to extract video features from the video demonstration, $\mathbf{v}' = f_v(V)$, see Fig. 3 (top left). Our model supports either single-view or multi-view inputs. When available, multi-view observations of a skilled physical activity can give valuable detail; for example, an egocentric view of the basketball being shot by the hands along with one or more exocentric viewpoints of the player’s full body pose would provide complementary information. When using multiple views, $\mathbf{v}'$ is simply the concatenation of features from individual views. The video spans $\mu_1 + \mu_2$ seconds, as discussed in Sec. 3.3, and one feature is extracted per second. Therefore, $\mathbf{v}'$ is a vector of dimension $(\mu_1 + \mu_2) \times N$ where N is the output feature dimension.

Note that our representation is video-frame based, meaning the person’s body pose is visible in the RGB but not explicitly extracted. While one could alternatively supplement the input with explicit body poses, our early experiments showed that the accuracy advantage is minimal compared to the high compute needed to get state-of-the-art poses [79], i.e., 34 seconds for a 10 second clip.

Pretraining: Multimodal LLM for skill-attribute generation. After encoding the videos, we input it to a multimodal LLM (M-LLM), denoted as $\mathcal{L}$. An LLM is a suitable choice since the output attributes are expressed in text format. We prompt the model as follows: “<video> Here is a video of a person doing <sport name>. Highlight up to <k> key concept areas where the person can improve: ...”. We replace the ‘<video>’ tag with video encodings obtained above.

The output of the M-LLM contains ‘ k ’ skill-attributes, distinct phrases that represent the dimensions of suboptimal execution in the provided video demonstration. These outputs are parsed to obtain the skill-attributes set $\hat{S}$, see Fig. 3 (bottom left). To this end, we employ a trainable mapper f_m that converts the feature vector $\mathbf{v}'$ to a representation compatible with the LLM, $\mathbf{v} = f_m(\mathbf{v}')$. The idea is that f_v is a large pretrained model and kept frozen, while f_m is trainable to convert the visual features to a multimodal representation. The supervising signal to train this M-LLM is the skill-attribute set S obtained in Sec. 3.3. We use standard log-likelihood loss [30, 53, 63, 86]. We emphasize that this pretraining enables the model to *generate* skill-attributes, as opposed to retrieving from a closed set. We show the superiority of this approach over retrieval in the experiments.

This pretraining stage results in aligning the video features $\mathbf{v}$ towards skill-attributes, thus making it suitable for the other axes of assessment—actionable feedback and proficiency estimation. We now use the pretraining weights for completing the assessment task suite, as described next.

Skill-attributes for actionable feedback. For generating expert actionable feedback, we provide the output skill-attributes $\hat{S}$, along with video V , as input to the model $\mathcal{L}$ and output the actionable feedback. See Fig. 3 (bottom middle). Next, we prompt the model $\mathcal{L}$ with a prompt as follows: “<video> Here is a video of a person doing <sport name>. Here are some possible axes that need improvement, as rated by an AI coach (may contain mistakes): < $\hat{S}$ >. Give feedback on the execution that will help the person improve.” As above, <video> is replaced by $\mathbf{v}$. Importantly, this prompt provides both V and $\hat{S}$ as an additional guiding signal for the actionable feedback generation process. The generated actionable feedback T contains ideas for improvement that are personalized to the video specifics, e.g., “you need to bend more while dribbling to maintain control” as opposed to simply naming the skill dimension to improve, e.g., *control*. While skill-attributes can be the same for two sports, the resulting actionable feedback will be geared towards the specific sport. This factoring helps make our model transferable, while also enabling sport-specific actionable feedback.

Skill-attributes for proficiency estimation. Lastly, we deploy skill-attributes for proficiency estimation. As we want to discern how the skill-attributes representation compares to standard video features without such training, we employ a linear probe setting where we have a linear layer f_p that takes in frozen $\mathbf{v}$ and outputs a class representing the proficiency P , see Fig. 3 (bottom right). The proficiency is a label of expertise: *novice*, *intermediate*, *expert*, or *late expert*. Only f_p is trainable.

3.5 Training and inference settings and implementation details

Train/test splits. Our approach aims to improve both traditional in-domain and zero-shot skill assessment. The datasets organize video clips by their superclass sport and their subclass skill. A skill is a drill or specific exercise, and each sport can have multiple skills. Ego-Exo4D has 3

Table 1: **Quantitative results.** Skill-attribute generation results (IoU@0.7) for Ego-Exo4D [37] and QEVD [68] (top left). Actionable feedback generation results for Ego-Exo4D [37] (top right) and QEVD [68] (bottom left). Metrics used to match SOTA on the corresponding dataset, B@4=BLEU@4, M=Meteor, R-L=ROUGE-L, B=BERT score. Proficiency estimation for individual sports (bottom right). Standard errors are reported in text.

Skill-attribute generation			Actionable feedback on Ego-Exo4D [37]				
Method	IoU@0.7	Ego-Exo4D [37]	QEVD [68]	Method	B@4	M	R-L
InternVideo2-NN [88]		14.0	23.8	InternVideo2-NN [88]	42.1	46.9	49.3
InternVideo2-FT [88]		15.0	24.5	InternVideo2-FT [88]	42.9	47.6	50.0
VideoChat2 [47]		9.3	16.9	VideoChat2 [47]	27.8	44.3	41.9
LLaVA [53]		9.7	17.3	LLaVA [53]	28.5	44.1	44.2
LLaVA-FT [53]		14.6	26.9	LLaVA-FT [53]	43.5	48.5	51.5
Stream-VLM [68]		14.5	28.0	LLaVA-FT w/ pose [53]	43.6	48.5	51.7
ExpertAF [9]		15.0	28.1	PoseScript/Fix [23, 24]	24.1	44.5	46.3
Attribute-Retrieval		19.7	32.4	ExpertAF [9, 63]	44.9	49.6	54.6
CROSSTRAINER		25.7	37.6	CROSSTRAINER	45.6	51.7	57.8
				w/o two-stage	43.8	48.8	52.3

Actionable feedback on QEVD [68]			
Method	M	R-L	B
Socratic-LLaMA-2-7B	9.4	7.1	86.0
Video-ChatGPT [56]	10.8	9.3	86.3
LLaMA-VID [50]	10.6	9.0	86.0
Stream-VLM [68]	12.7	11.2	86.3
CROSSTRAINER	17.6	18.1	87.8
w/o two-stage	12.1	10.8	86.0

Proficiency estimation on Ego-Exo4D [37]			
Method	B.ball	Soccer	Rock Cl.
EgoVLPv2 [74]	48.0	62.5	34.0
CROSSTRAINER	53.1	68.8	37.1

superclasses and 5 subclasses: soccer has skills dribbling and penalty kick; basketball has skills Mikan layup, reverse layup, jump shot; rock climbing is not sorted into skills. QEVD has 1 superclass (fitness) and 23 subclass skills (jumping jacks, squats, etc.). To explore models’ generalization ability, we perform controlled experiments with the following train/test settings, in decreasing volume of available training data (see Fig. 3 (right)):

- *Fully supervised (FS)*: Train on all sports and skills, and test on held-out set of videos. This represents in-domain testing.
- *All sport zero-shot (ZS-1)*: Train on all sports and skills, *except* the target skill. This means other skills from the same sport are seen during training.
- *Familiar sport zero-shot (ZS-2)*: Train on all skills from the same sport, *except* the target skill. This means only the same sport is seen during training.
- *Novel sport zero-shot (ZS-3)*: Train only on $n - 1$ sports, test on skills from the n -th unseen sport. This means other skills from the same sport are *not* seen during training.

For each controlled experiment, we retrain the model to avoid any information leak between the train and the test splits, and between the seen and novel sports.

Model architecture. f_v is EgoVLPv2 [74] for Ego-Exo4D [37] (precomputed with dataset) and CLIP [75] for QEVD [68] (lightweight to extract). We use one feature per second with a 4096-d output. For Ego-Exo4D [37], we use the ego and four exo views, while QEVD [68] is single-view. f_m is a two layered MLP with GELU activation, consistent with [52]. The MLP+GELU module takes in 4096-d representation, consistent with the embedding dimension of the multimodal LLM $\mathcal{L}$, Llama-3.1-8B-Instruct [5]. f_p is a linear layer with output size 4, the number of proficiency levels.

Training details. We train both $\mathcal{F}_a$ and $\mathcal{F}_t$ in LoRA setting [38], with rank 128, alpha 256, and dropout 0.05, for efficiency. The best performance is obtained with a learning rate of 2×10^{-3} for f_m and 2×10^{-4} for $\mathcal{L}$. Recall that f_v is kept frozen. The model is trained for 2 epochs or till convergence. Total training time depends on the dataset setting, varying between 1 – 3 hours. All experiments are performed on one GH200 NVIDIA node.

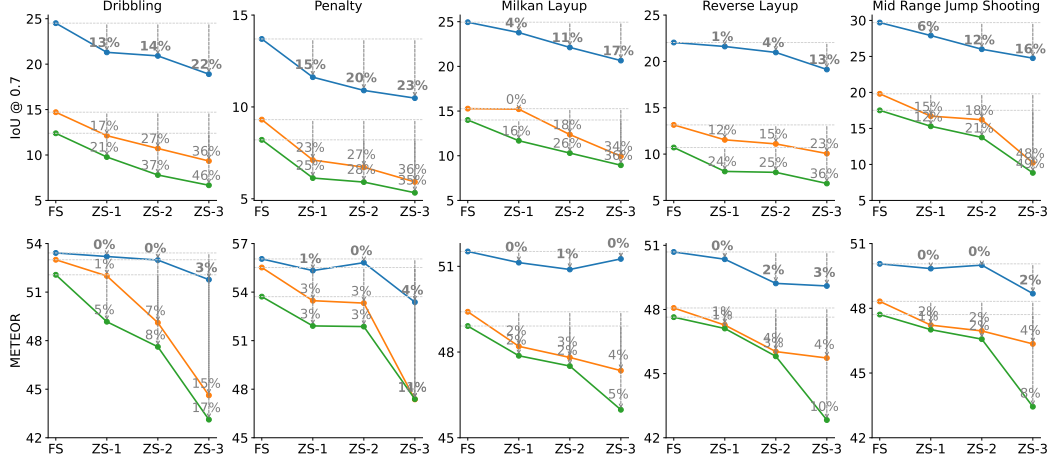


Figure 4: **Zero-shot performance.** Performance trend when testing on various skills (dribbling, penalty, etc.) for different in-domain and zero-shot training settings (FS, ZS-1, etc.) for skill-attribute generation (top) and actionable feedback generation (bottom) for Ego-Exo4D. The relative drop in performance w.r.t. FS is shown as a percentage. Our method is consistently the best for all methods, and the relative drop is the least for all zero-shot variants (ideal curve would be flat and high). See Supp. for QEVD. Legend: — CROSSTRAINER, — ExpertAF [9] and — InternVideo2 [88].

4 Experiments and Results

We first present actionable feedback generation (Sec. 4.1) for its focus in prior work and directly comparable SOTA methods. Next, we validate the skill-attributes pretraining (Sec. 4.2). Finally, we present proficiency estimation (Sec. 4.3) and explore long-tail in-the-wild YouTube videos (Sec. 4.4).

4.1 Generating actionable feedback

First we evaluate actionable feedback generation: given a video, generate the natural language commentary explaining what to correct.

Baselines. We compare against the SOTA ExpertAF [9] and Stream-VLM [68] models as well as the original baselines from their respective experiments, which include strong multimodal LLMs with model size 7B-8B. CROSSTRAINER has fewer trainable parameters due to our use of LoRA [38], but we leave the baselines in their full (non-LoRA) form to report their most accurate numbers. “w/o two-stage” is the end-to-end ablation not using skill-attributes.

Metrics. Following [9, 68], we report language metrics—METEOR [11], BLEU-4 [69], ROUGE [78] and BERT-score [25], reported out of 100; higher is better. While establishing ground truth commentary is nuanced and there could be multiple valid feedback statements for a given video, prior work [9] shows that this evaluation paradigm correlates strongly with human subjects’ evaluation of the generated commentary.

Fully supervised in-domain results. Tab. 1 (top right and bottom left) shows the results. CROSSTRAINER clearly outperforms all prior work and strong baselines. Standard error is less than 0.1 for all metrics. This result clearly supports using skill-attributes as an intermediate representation for actionable feedback. We investigate the robustness of actionable feedback generation to the choice of the LLM $\mathcal{L}$ and the noise level in the Supp. Fig. 5 (first two rows) shows qualitative outputs for both datasets, where our model correctly captures the expert’s feedback.

Zero-shot transfer. Fig. 4 (bottom row) shows the trend when transferring the knowledge from one domain to another in Ego-Exo4D [37] (see Supp. for QEVD [68]; results are similar). We plot the best M-LLM (ExpertAF [9]) and retrieval (InternVideo2-FT [88]) baselines; we see a similar trend in all other baselines. Our method obtains the best performance for all training settings and degrades most gracefully as training data coverage diminishes in the increasingly difficult zero-shot settings (ZS-1, ZS-2, ZS-3). Our max drop is 4%, vs. 17% for the baseline. In Fig. 5 (bottom left), we show a confusion matrix denoting better transfer from soccer to basketball, and vice versa,

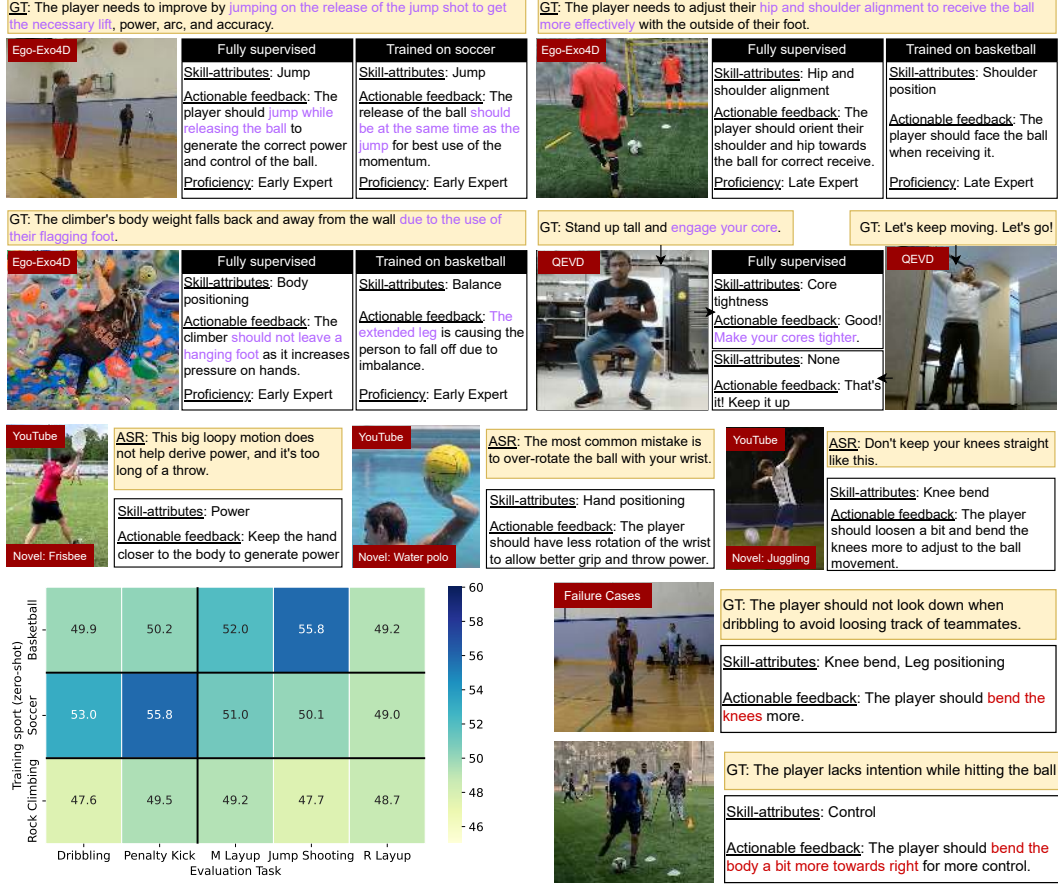


Figure 5: **Qualitative results.** CROSSTRAINER generates skill-attributes, actionable feedback, and proficiency for samples from both Ego-Exo4D [37] and QEVD [68]. The outputs are meaningful even in the zero-shot setting (first two rows). Our method is also applied to in-the-wild videos from YouTube with novel sports (frisbee and water polo) and even new drills (juggling in soccer) with feedback matching the YouTube expert’s comments (transcribed with ASR) (third row). Confusion matrix shows better transfer between related sports (bottom left). **Failure cases** here and in Supp. show the difficulty of the task, especially non-visual feedback like *lacking intent* (bottom right).

compared to rock climbing—reinforcing observations in cognitive science [43, 65, 81]. In summary, CROSSTRAINER demonstrates robust to transfer to novel sports and drills.

4.2 Learning skill-attributes in pretraining

Next we evaluate the skill-attribute prediction (c.f. Sec. 3.3).

Baselines. We compare to retrieval baselines (InternVideo2 [88], nearest neighbor (NN) and finetuned (FT), and Attribute-Retrieval trained with contrastive learning), zero-shot multimodal baselines (VideoChat2 [47], LLaVA [53]), and a finetuned version of the best multimodal baseline (LLaVA-FT). In addition, we again compare to ExpertAF [9], a SOTA model for generating actionable feedback, but here we convert its output to skill attributes by text-only LLM prompting [2]. All the baselines have access to the same actionable feedback data as ours [37, 68], and they directly supervise with it.

Metrics. Since skill-attributes are a list of phrases, we employ IoU-based matching (a.k.a. Jaccard index) [21, 31] between the inferred annotations (Sec. 3.3) and generated skill-attributes. We call a pair a match if BERT-score is more than k , and report $\text{IoU}@k$ for $k=0.7$ here and $\{0.8, 1.0\}$ in Supp.

Fully supervised in-domain results. Tab. 1 (top left) shows the results for Ego-Exo4D [37] and QEVD [68]. CROSSTRAINER outperforms all zero-shot and trained baselines that *post process* actionable feedback to obtain skill-attributes, by more than 10%. Moreover, our method outperforms Attribute-Retrieval by 6%—showcasing the effectiveness of our generative approach. The standard

error is less than 0.5 for all metrics and all methods. Fig. 5 shows examples predicting the skill-attributes that the person should improve.

Zero-shot transfer. Fig. 4 (top row of plots) shows the zero-shot transfer results for Ego-Exo4D [37] (see Supp. for QEVD [68]; results are similar). As above, we see a clear advantage of our training scheme. Moreover, CROSSTRAINER declines much more gracefully than the baselines. Overall, these results show the effectiveness of our skill-attribute guided pretraining for zero-shot transfer.

4.3 Estimating demonstrator proficiency

Next we evaluate the quality of the learned video representation $\mathbf{v}$ for proficiency estimation.

Baselines and metrics. We compare the strength of the video representation with respect to the frozen representation f_v , i.e., EgoVLPv2 [74] for all scenarios. We report classification accuracy.

Results. Tab. 1 (bottom right) shows the results. We see a clear gain of up to 6% when using the pretrained video representation, with standard error < 1 . Fig. 5 shows some example predictions, where our model can distinguish between *early* and *late expert*.² This experiment suggests another potential use case in pretraining skill-centric representations for better assessment and feedback.

4.4 Testing long tail in-the-wild YouTube sports videos

Finally, we explore applying CROSSTRAINER to novel sports and drills in in-the-wild videos from YouTube. We extract 10 clips from tutorial videos where an expert tutor demonstrates the incorrect way of doing a drill, while also explaining the incorrectness (i.e., the target feedback, which is withheld from our model). Details of obtaining the videos from YouTube are given in the Supp. We use the expert’s transcribed ASR text to compare with the output of our model.

Results. Fig. 5 (fourth row) shows example results (rest in Supp.). We see that a model trained on Ego-Exo4D [37] videos (soccer, basketball, rock climbing), is able to predict the issue with demonstrations in novel long-tail sports frisbee and water polo. Moreover, we also see correct feedback in a novel juggling drill in soccer, which CROSSTRAINER is not trained with. We observe that learning a sport-specific vocabulary is difficult in zero-shot transfer. Nonetheless, the essence is captured and described in text, e.g., even though the model does not understand the *loopy motion*, it understands that keeping the hands closer to the body helps in generating more power—a fact known and applied in various physical scenarios. We also verify the generations with a user study. Human subjects not associated with the project rated 75% of the generations as actionable and correct. Every generation is judged by three raters, and we take a majority vote.

Failure cases in Fig. 5 (bottom right) showcase the difficulty in capturing subtle mistakes, especially non-visual feedback like *intent*, *decision*. We further discuss limitations and societal impact in Supp.

5 Conclusion

We introduced CROSSTRAINER—a novel approach that discovers *skill-attributes* from video demonstrations. These skill-attributes are generalizable, enabling a zero-shot transfer of skill assessment to novel sports and drills. Our experiments show notable gains in actionable feedback generation and proficiency estimation for both in-domain and zero-shot settings. In the future, we will explore ways to quantify sport relatedness to predict the transferability between sports, as well as explicit representations to capture the environmental context.

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²Note that QEVD does not provide proficiency labels.

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A Supplementary material for Learning Skill-Attributes for Transferable Assessment in Video

A.1 Table of content

This supplementary contains the following:

- **A supplementary video** that first motivates the problem with some example actionable feedback requests from learners on Reddit (discussed in Sec. 3.3). Next, we show qualitative results from Ego-Exo4D [37], QEVD [68], and YouTube videos. Finally, we also show some failure cases.
- **Sec. A.2: LLM prompt** that we use to obtain skill-commentary from expert commentary.
- **Sec. A.4: Additional quantitative** results to show metrics for skill-attribute generation, and QEVD [68] zero-shot results.
- We discuss **limitations** in Sec. A.5.
- We also discuss **societal impact** in Sec. A.6.

A.2 LLM prompt for obtaining skill-attribute from expert commentary

In Sec. 3.3, we discuss using a large language model (LLM) to extract skill-attributes that are suboptimally performed. We use the following prompt:

System: Answer the question regarding a commentary about a sports drill. Do not add information not present in the question.
User: The transcript of an expert commentating on a SPORT NAME COMES HERE drill is given below. List down the concepts that are correct and incorrect in the drill, as noted by the expert. The concepts are distinct aspects of the skill, e.g., control, body positioning, speed, body movement, hand position, and so on. Feel free to come up with newer concepts and write the response in two lines. The first line should contain the correctly shown concepts, and the second line should contain the incorrectly shown concepts. It should be in this format.
Correct - comma separated concepts.
Incorrect - comma separated concepts.
Here is the expert feedback:
NARRATION COMES HERE
Assistant:

We prompt the LLM to provide both correctly and incorrectly demonstrated skill-attributes. Our Attribute-Retrieval baseline (Sec. 4.3) uses both of them.

A.3 Obtaining YouTube videos for testing long tail in-the-wild sports and drills

We evaluate the performance of our model on zero-shot sports and drills from in-the-wild YouTube videos in Sec. 4.4. To obtain a dataset for this test, we create a list of novel sports and novel drills within the Ego-Exo4D [37] and QEVD [68] sports (basketball, soccer, rock climbing, exercise), and search for videos on YouTube with their coaching videos. Our criteria is that the video should contain a mistake done by a learner, and a coach giving feedback. Note that many videos show only the correct demonstration; hence, obtaining videos for our task is challenging.

Specifically, we randomly selected some common sports—soccer (juggling), basketball (dribbling) and some rarer ones like water polo, korfbal, polo, jai alai, kin ball, frisbee. We search more than 50 videos with keywords like “Coaching video of <sport/drill>”, “<sport/drill> training session for beginners”, “<sport/drill> common mistakes for beginners”, “Dos and don’ts in <sport/drill>”. The videos are manually watched to find the desired coaching instances. The process took overall 12 hours, and was done by graduate students not associated with this project.

Table 2: Buckets of exercises in QEVD [68] grouped by similarity in execution and effect.

Group name	Exercises	Remark
Stretches & mobility	quad stretch, armcrosschest, good morning beginner, floor touches, toe touchers	Focused on flexibility and range of motion; often used in warm-up or cooldown phases. Involves static or slow dynamic movement.
Cardio & agility	high knees, quick feet, jumping jacks, air jump rope, butt kickers, puddle jumps	Elevates heart rate with low to moderate resistance; emphasizes agility and coordination with repetitive footwork.
Leg strength & lower-body	squats, squat jumps, squat kicks, walking lunges, lunge jumps, standing kicks	Targets glutes, quads, hamstrings through controlled or explosive leg movements. Builds strength and endurance.
Core & upper-body	plank taps, moving plank, pushups, shoulder gators	Focuses on core stabilization and upper body strength, particularly arms, shoulders, and chest. Often bodyweight-based.
Full-body	boxing squat punches, mountain climbers	High-intensity, compound movements that engage multiple muscle groups while promoting coordination and rhythm.

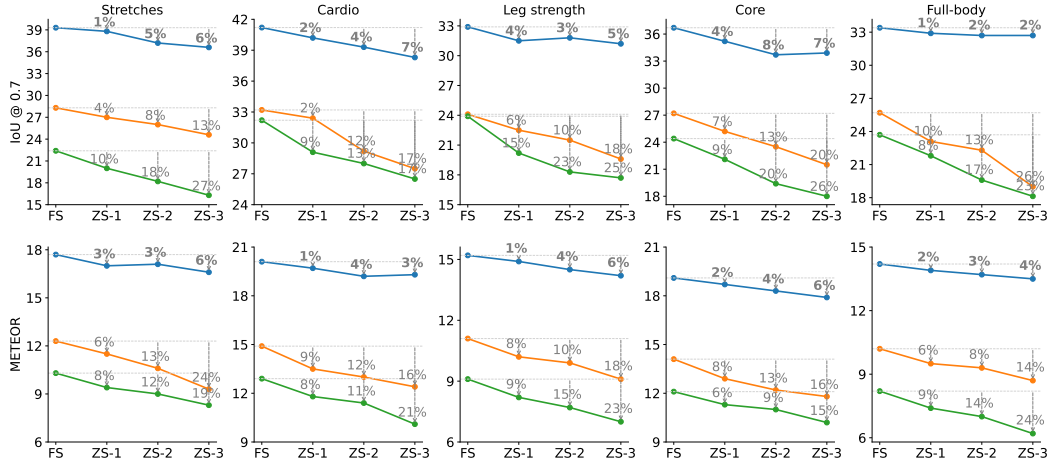


Figure 6: **Zero-shot performance.** Performance trend when testing on various skills (stretches, cardio, etc.) for different in-domain and zero-shot training settings (FS, ZS-1, etc.) for skill-attribute generation (top) and actionable feedback generation (bottom) for QEVD [68]. Legend: — CROSSTRAINER, — Stream-VLM [68] and — InternVideo2 [88].

A.4 Additional quantitative results

In Sec. 4.2 and Tab. 1 (top left), we show results for generating skill-attributes for IoU@0.7. We extend the table to show results on IoU@ k for $k \in \{0.7, 0.8, 1.0\}$ in Tab. 3.

Next, we show zero-shot transfer performance on the QEVD [68] dataset (summarized in main paper, and detailed here due to space limitations). Note that all videos in QEVD are for fitness exercises, and they are not as distinct as a different sport. Moreover, every video contains multiple exercises, with the transition labeled as “Moving to (EXERCISE NAME)...”. We use these labels to split the videos per-exercise. We discard instances that are before the start of any labeled exercise. Next, we create sport labels based on similarity in execution and effects. We create this division for the purpose of zero-shot transfer experiments. This split is created using consensus from ChatGPT-4o [2] and Llama-3.1 [5], and finally manually verified. The splits and the reasonings are given in Tab. 2. A total of 23 unique exercises are divided into 5 groups.

Table 3: **Quantitative results.** Skill-attribute generation results for IoU@ k for $k \in \{0.7, 0.8, 1.0\}$ for Ego-Exo4D [37] and QEVD [68], extension of Tab. 1 (top left) on remaining k values.

Method	IoU	@0.7	@0.8	@1.0	Method	IoU	@0.7	@0.8	@1.0
InternVideo2-NN [88]	14.0	8.9	7.7		InternVideo2-NN [88]	23.8	16.3	14.8	
InternVideo2-FT [88]	15.0	9.4	8.2		InternVideo2-FT [88]	24.5	16.6	15.3	
VideoChat2 [47]	9.3	6.6	4.0		VideoChat2 [47]	16.9	11.4	10.1	
LLaVA [53]	9.7	7.2	4.9		LLaVA [53]	17.3	12.5	11.8	
LLaVA-FT [53]	14.6	9.1	8.1		LLaVA-FT [53]	26.9	19.2	18.2	
Stream-VLM [68]	14.5	9.1	8.3		Stream-VLM [68]	28.0	19.9	18.6	
ExpertAF [9]	15.0	9.5	8.4		ExpertAF [9]	28.1	19.7	18.3	
Attribute-Retrieval	19.7	12.7	10.7		Attribute-Retrieval	32.4	24.7	23.0	
CROSSTRAINER	25.7	15.9	14.4		CROSSTRAINER	37.6	29.8	28.1	

Table 4: **Robustness and sensitivity of LLM:** Comparison of actionable feedback generation with various sources of skill-attributes (left). Comparison of BERT similarity score of skill-attributes generated by our method with skill-attributes obtained from different LLMs and prompts (right).

Skill-attrib. source	Test-set $\mathcal{L}$	B@4	M	R-L		
Llama-3 8B (orig)	Llama-3 8B (orig)	45.6	51.7	57.8	Ours vs	Score
Mistral 8B [4]	Llama-3 8B (orig)	45.3	51.8	57.5		
Llama-3 8B (orig)	Mistral 8B	45.8	51.8	57.8		
Llama-3 8B (orig)						
w/ 10% noise	Llama-3 8B (orig)	45.2	50.5	56.2	Llama-3 8B w/ prompt choice 1	0.99
w/ 20% noise	Llama-3 8B (orig)	45.1	50.0	54.9	Llama-3 8B w/ prompt choice 2	0.98
w/ 30% noise	Llama-3 8B (orig)	44.3	49.4	53.1	Mistral 8B [4]	0.98
w/ 50% noise	Llama-3 8B (orig)	42.7	47.3	50.2		
w/ 70% noise	Llama-3 8B (orig)	41.9	46.3	49.6		

Fig. 6 shows the results. First, in skill-attribute generation (top row), we see that our method outperforms both Stream-VLM [68] and InternVideo2 [88] baselines. Moreover, as seen in Ego-Exo4D [37], the performance decrease in the zero-shot setting is milder than the drop observed in the baselines. Finally, we see a similar trend in actionable feedback generation (bottom row). Overall, zero-shot results in both Ego-Exo4D [37] and QEVD [68] show that our idea of learning to generalize using skill-attributes is effective.

Finally, to show the robustness and sensitivity, we perform the following experiments on Ego-Exo4D [37] actionable feedback generation:

Robustness to the choice of the language model: We train and evaluate skill-attributes using different language models. We first train the model using Mistral’s 8B language model (mistralai/Minstral-8B-Instruct-2410) [4] and compare it against the skill-attributes test set obtained using Llama-3.1-8B-Instruct, and vice-versa. Tab. 4 (left) shows the results. We see that the model is robust to the choice of the language model, and using any strong language model helps achieve a good performance.

Actionable feedback generation w/ noisy skill-attributes: We inject noise in the actionable feedback generation evaluation. We replace $X\%$ of inferred skill-attributes with a random skill-attribute and observe the performance at various levels of noise X . See results in the table below. We observe that adding noise degrades the performance, with the performance matching that of end-to-end direct training at $X = 20\%$. We can conclude that the performance is positively correlated to the quality of the generated skill-attribute. Improving that will also improve the actionable feedback performance. These ablation studies further showcase the effectiveness of using skill-attributes for actionable feedback.

Correlation between skill-attributes generated using different LLMs: We try two prompt variants, and a different language model (Mistral 8B, mistralai/Minstral-8B-Instruct-2410) [4] to compare the similarity between generated skill-attributes—checking if our idea is independent of the chosen language model. We use Hungarian matching to find the most-similar match between the new skill-attribute set and our original skill-attribute set. Next, we find the average BERT score [25] between the sets. Tab 4 (right) shows the results. We see a very high similarity between skill-attributes

generated from different prompts, and a different language model. This result implies that our idea is independent of the choice of a reasonable language model.

A.5 Limitations

We observe that the proposed model struggles with feedback that is about aspects not directly visible in the video. Phrases like *lacking intent* is not groundable definitively and hence, is not captured. Nevertheless, this inability to capture abstract notions is also observable in all the baselines, and in general, vision encoders. Secondly, as we discuss in Sec. 4.1, commentary is subjective and there can be various correct ways of providing feedback that improves a learner’s performance.

Moreover, in this work, we do not factor aspects like terrain and opponent behavior. This assumption is consistent with the datasets Ego-Exo4D [37] and QEVD [68] that are both single-person, and do not consider external factors. Furthermore, prior work also considers single-person skill assessment/feedback [9, 12, 37]. There are research works in multi-agent cooperation, but they are restricted to simulation and simpler objectives than performance feedback.

A.6 Societal impact

Our CROSSTRAINER can be used for learning skills, especially long-tailed low-resource sports like *kho-kho*, *shinty*. On the positive side, our model democratizes access to skill coaching, and it promotes inclusivity in underrepresented sports. More learning will promote more people playing the sport, and eventually, more data for training and expansion of knowledge. However, the model is trained with Ego-Exo4D [37] and QEVD [68] that might have regional bias. We believe as more data is available, the biases will go down, and we will move closer towards full physical skill understanding.

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